



Cooler Buildings for **Hotter** Climates.

We are an international community initiative, researching, designing, developing and sharing building solutions for climate-vulnerable countries.



"My room feels much comfortable especially the ventilation and the view. When I grow up I will build such a house. I love my house."

- Girl living at the Al-Mizan Children's Ecovillage

98.8%

of monitored hours across four bedrooms were within a healthy temperature range without air conditioning, throughout 2024*

*benchmarked against international comfort standards adapted for tropical humidity.

A message from Andrew Simmonds, Team Lead

For 20 years, many of you and I built the AECB and Passivhaus network together - a UK-wide community of building physics and carbon-literate designers, builders and suppliers. This community has thrived, sharing knowledge, expertise, optimism, and goodwill through the most challenging times.

In recent years, some members have applied their expertise to building designs in less wealthy, climate-vulnerable, hot and humid countries: researching, designing, building and sharing results. ARC continues that work, improving thermal safety, health and wellbeing for communities beyond the UK who urgently need it.

I invite you to support ARC's action research: Can we build effective, resilient homes that stay liveable through extreme heat without complex technologies people can't afford? How safe can naturally ventilated buildings stay as the climate destabilises and extreme weather increases? How well do passive-only measures work: orientation, reflectivity, shading, cross ventilation? Is 'multi-mode' energy efficient performance practical: 'free running' sometimes, switching to mechanically cooled, dehumidified and ventilated during periods of extreme heat? How safe are building designs when the power fails during heatwaves?

At Islamic Help's Al-Mizan Children's Ecovillage in Tanzania, our flagship building has held safe, healthy indoor conditions for 98.8% of monitored hours (2024), with no air conditioning. What particularly inspires us is the strong working relationships developing with our international friends and colleagues, such as the committed and knowledgeable staff.

Thank you to everyone who has already supported ARC - your backing has made this possible. I hope more of you will join us.



Team Lead, ARC



Meet our ARC People



Huda Elsherif

Building physics engineer and certified Passivhaus designer. I'm passionate about thermal comfort and teach at the University of Kent.



Archie Worthington

Architecture Design Studies student in Liverpool and R&D Associate at ARC.



Kamran Shezad

Faith-led environmentalist linking ARC's work to communities.



Toby Cambray

Building physicist specialising in moisture simulation, monitoring and analysis, as well as Passivhaus and retrofit.



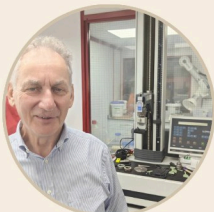
Suleiman Dauda

School teacher with 13 years' experience at a Vikindu state-funded secondary school, Tanzania.



Marwa Mohabe

Environmental policy professional from Tanzania focused on waste management, circular economy, and climate resilience.



Eric Rirsch

Ph.D qualified Physical Chemist working for Safeguard Europe; a company developing products to control and manage moisture problems in buildings.



Hudson Lambert

Managing Director at Safeguard Europe Ltd, where I've spent the past 30 years working on practical solutions to protect buildings from the elements.



Adele Mills

Registered UK architect, qualified 1989. Set up ecological design/build practice Simmonds.Mills in 1992.



Ben Roberts

Building energy lecturer researching heat, air quality, and comfort, mainly in tropical and low-resource settings, using fieldwork and modelling.



Shely Begum

Associate Architect at AtkinsRéalis. I specialise in environmentally responsible buildings with hands-on site work in the UK and rural Bangladesh.



Innocent Peter

Civil & Structural Engineer specializing in building design, construction, infrastructure, and geotechnical engineering.

Who we are

We started Architecture for Resilient Communities to explore and share evidence-based, practical ways to keep buildings safe and liveable as the climate heats up. We collaborate with communities, engineers, designers, builders, and environmental champions working in hot and humid regions where the conventional approach to cooling, air conditioning, is unaffordable and unreliable. ARC is developing a technical and design framework, and a Cool Buildings performance standard based on an affordable, adaptive-comfort approach. We aim to publish what we learn free of charge for all, particularly for those communities most vulnerable to climate change.

The problem

For most people in climate-vulnerable regions, reliable air conditioning is out of reach. Costly to install, costly to run, often inefficient, unhealthy and dependent on a power grid that often fails, especially during heatwaves. Designers and builders face a second problem: There is little measured data on how buildings in these climates actually perform, so they are often 'flying blind'. Methods that were adequate a generation ago may no longer be safe in our rapidly changing climate.

Our approach

The ARC Cool Buildings Standard helps designers and builders minimise overheating in hot and humid climates, for places where reaching a standard like Passivhaus in one step is unrealistic. The strategy rests on a 'two-step' improvement pathway, and a 'three-zone' room planning method: build first for natural ventilation and passive cooling, and design so that mechanical cooling, dehumidification and ventilation can be added later, room by room, as budgets allow and climate demands. Passive shading and cooling, enhanced with low-power technologies such as fans, reduce reliance on electricity for occupants' thermal safety. The standard also requires a communal heat refuge operating safely during extreme heat and power cuts.

What we learn from design and development, following monitoring and feedback, will become a free pattern library of design solutions, building-science resources and training materials, co-created with international experts and the communities who will use them.



Some of our work



Innovative wet and dry composting system, **يُخْرِجُ الْحَيَّ مِنَ الْمَيِّتِ**.



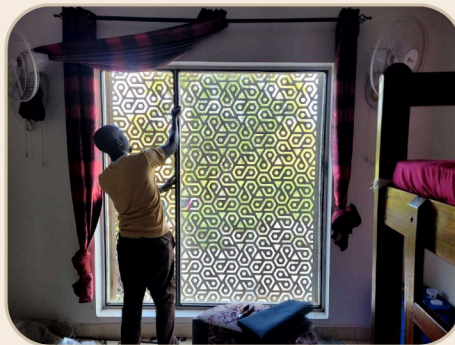
Trees for landscape cooling, food & nature, **قَة جَارِيَة**.



ARC weather station, **آيَات**.



Local fabrication of ARC patterns, **نَقَان**.



Secure, insect protected cross ventilation, **لَا ضَرَر وَلَا نَفْعَ**.



Coconut fibre-based material experiments, **إِحْتِيَاء**.



Thin masonry ferrocement modular ceiling, **فَقْدَرُهُ تَقْدِيرًا**.

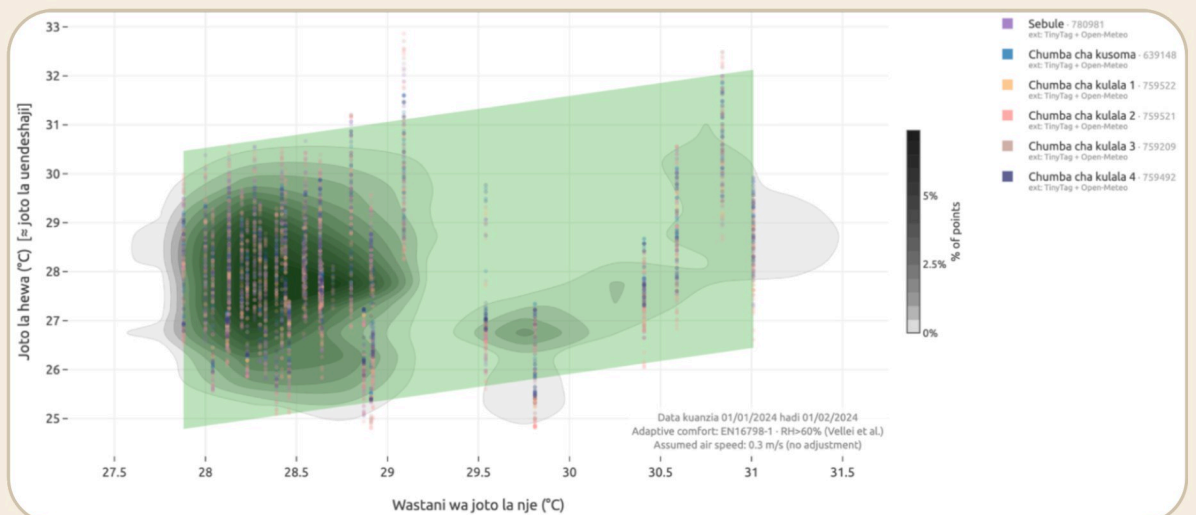


Structural experiments, **تَبَيَّنُوا**.



Self-build cool-roof construction, **وَتَعَاوَنُوا عَلَى الْيَرِّ**.

Adaptive comfort graph plotting indoor operative/air temperature for six of the rooms in House 5 in January 2024 against a weighted running mean of external temperatures. These points are plotted against an adaptive comfort band derived from a humidity-adjusted model created by Vellei et al. (2017). Assumed air speed of 0.3 m/s.



In their own words...

"My room feels much comfortable especially the ventilation and the view. When I grow up I will build such a house. I love my house."

- Girl living at the Al-Mizan Children's Ecovillage, whilst rating her current feeling as "Cold", recorded at 28°C, 75% humidity

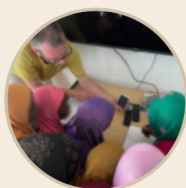
"I feel much safer in my house more than at home, I love my room mostly, because of the good weather, big windows, nice bed and fresh air."

- Girl living at the Al-Mizan Children's Ecovillage

"I felt good and I was in a good mood"

- Girl living at the Al-Mizan Children's Ecovillage, whilst rating her current feeling as "Cold", recorded at 25°C, 73% humidity

Our work so far, in numbers



2023

Monitored continuously since



1

weather station installed



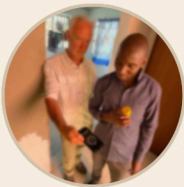
5

buildings being retrofitted



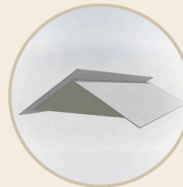
8

countries in our expert network



3

houses monitored (and 1 planned)



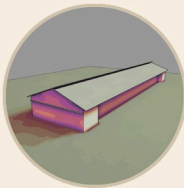
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ARC Cool Building Patterns developed



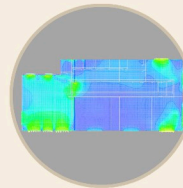
2

hot readings out of 70 in a thermal survey of occupants



2

buildings currently being designed



3

buildings analysed for performance



99%

of hours spent in healthy conditions without AC*



2

buildings designed and built



6

research and development components



Core funding and special projects

ARC runs a small, lean team harnessing a large amount of donated expertise. Much of what ARC delivers each year comes from our advisors, giving their time and resources for free. We treat all funding we receive as unrestricted - not earmarked for specific items, and we ask supporters to trust us to allocate funds where they're needed most. In practice, that means splitting the budget two ways.

Core funding keeps ARC running, paying for networking, research, monitoring and analysis, ARC pattern design and technical development, special project development and of course basic coordination and administration activities. ARC also collaborates in live projects as test beds for its R&D, involving research and development coordination - turning monitoring data, detailed investigations, and occupant feedback into design guidance.



Amanah (أمانة) - Trust, Faithfulness.

ARC is a highly collaborative venture, strategic direction and priorities are strongly informed through thoughtful discussion and debate with our expert advisors and international colleagues

Special projects require a higher level of funding to deliver and are supported with technical direction from the ARC team as well as legal, architectural and financial support. Special projects being developed include:



A demonstration retrofit

Upgrading an existing Tanzanian house to the ARC Cool Buildings Standard - to be used as a training hub, community rooms & accommodation.



ARC Contractors Club

Working with local builders and their clients to co-create and evidence cool-building techniques and patterns via workshops, sharing performance data: creating mutually supportive local construction ecosystems.



Low-cost DIY infrared sensor array & network

Developing a DIY kit based on affordable components, locally assembled, to measure humidity and air, surface and mean radiant temperatures.



Installing weather stations and data loggers

Expanding the evidence base for better building performance in hot climates and for educational use.



ARC Cool Building Patterns

Producing proof of concept designs, components, construction assemblies and 'ready to use patterns' for building in hot climates.



Knowledge sharing

Making Cool Buildings knowledge and ARC's pattern library accessible to students, contractors and their customers - and learning easier for all.

Thank you

Our sincere thanks go to the early supporters of this initiative, including the Islamic Help team and trustees, their regional office teams across Pakistan and Tanzania, and Simmonds.Mills Architects, the staff, residents, and neighbours of the Al-Mizan Children's Ecovillage in Tanzania, the AECB, and, of course, our volunteer expert advisors.

We are grateful to many people who have shared their home, their observations and their honest opinions with us, sharing their personal experiences, local knowledge and technical expertise.

Please consider supporting ARC, donations large and small are meaningful. If you'd like to donate, please visit actionresearchprojects.net/support

